# Analog Transmission

To **send the digital data over an analog media,** it needs to be **converted into analog signal.** There can be **two cases** according to data formatting.

**Bandpass:** The filters are used to **filter and pass frequencies of interest.** A bandpass is a band of frequencies which can pass the filter.

**Low-pass:** Low-pass is a **filter that passes low frequencies signals.**

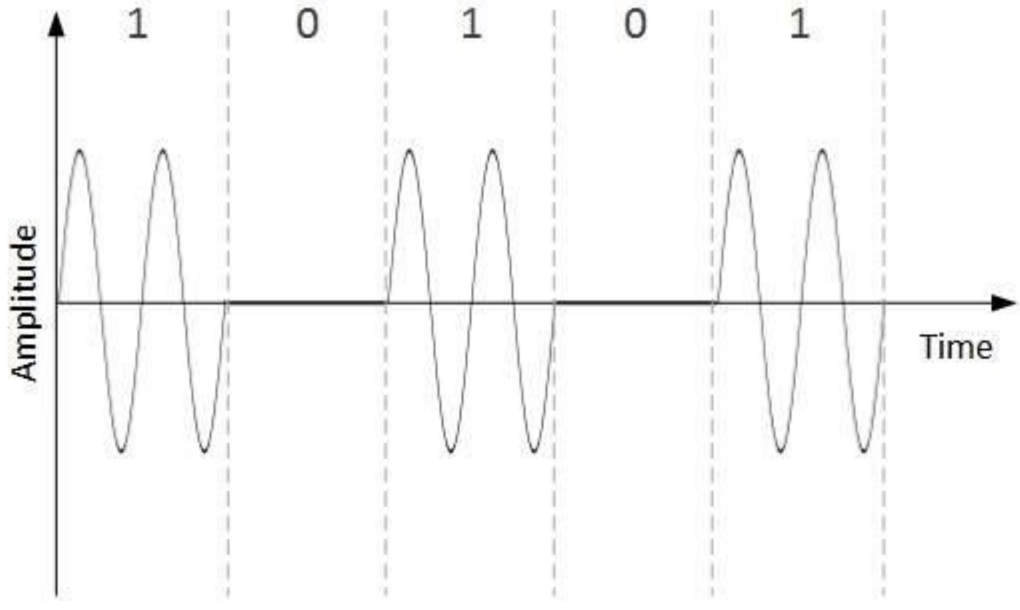
When **digital data** is **converted into a bandpass analog signal,** it is called **digital-to-analog conversion.** When **low-pass analog signal is converted into bandpass analog signal,** it is called **analog-to-analog conversion.**

# Digital-to-Analog Conversion

When **data** from one computer is **sent** to another **via some analog carrier,** it is first **converted into analog signals.** Analog signals are modified to reflect digital data. An **analog signal** is **characterized by its amplitude, frequency, and phase.** Therefore, there are **three kinds of digital-to-analog conversions.**

## Amplitude Shift Keying

In this conversion technique, the **amplitude of analog carrier signal is modified to reflect binary data.**




When **binary data represents digit 1,** the **amplitude is held; otherwise,** it is **set to 0.** Both **frequency and phase remain same** as in the original carrier signal.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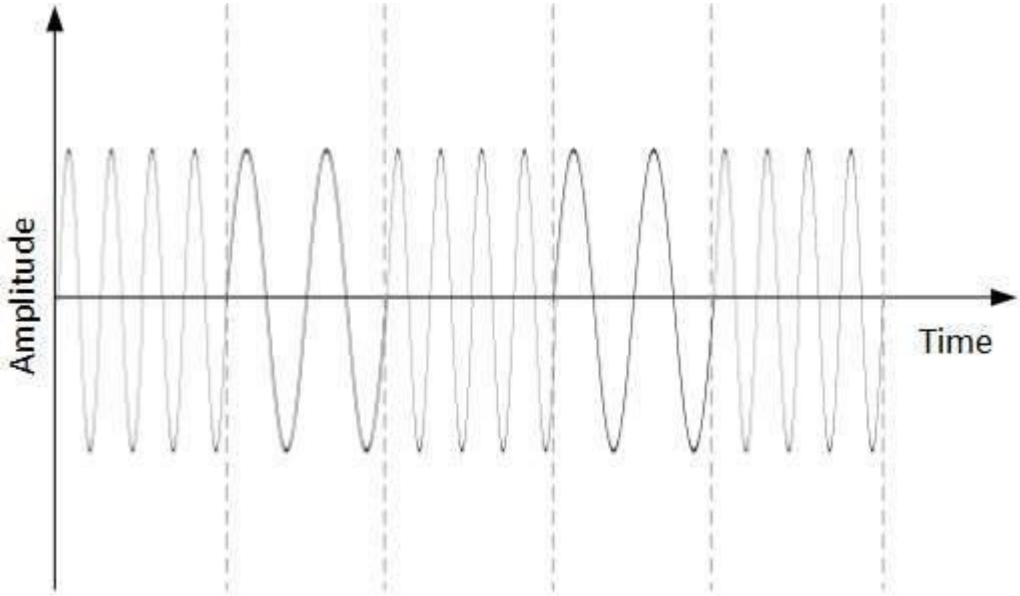
## Frequency Shift Keying

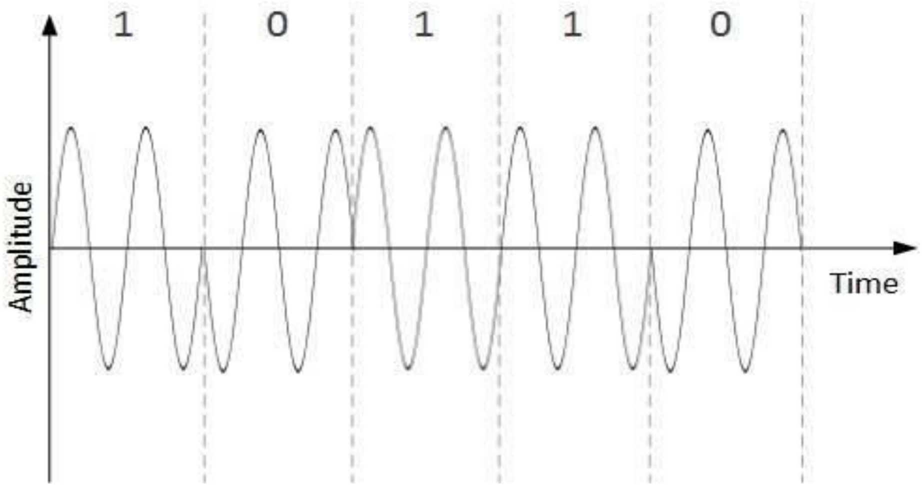
In this conversion technique, the **frequency of the analog carrier signal is modified to reflect binary data.**




This technique **uses two frequencies,** f1 and f2. One of them, for example **f1, is chosen to represent binary digit 1** and the **other one is used to represent binary digit 0.** Both **amplitude and phase** of the carrier wave are **kept intact.**

## Phase Shift Keying

In this conversion scheme, the **phase of the original carrier signal is altered to reflect the binary data.**




When a **new binary symbol is encountered,** the **phase of the signal is altered. Amplitude and frequency** of the original carrier signal is **kept intact.**

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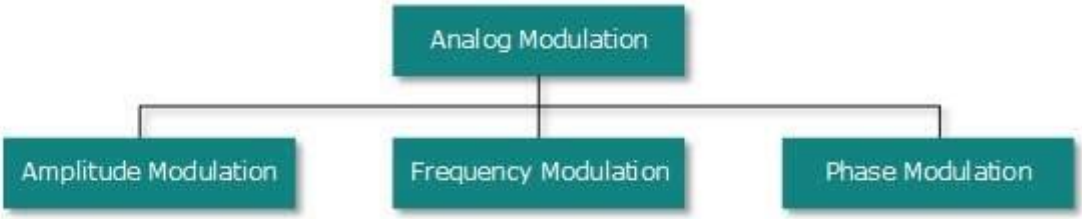
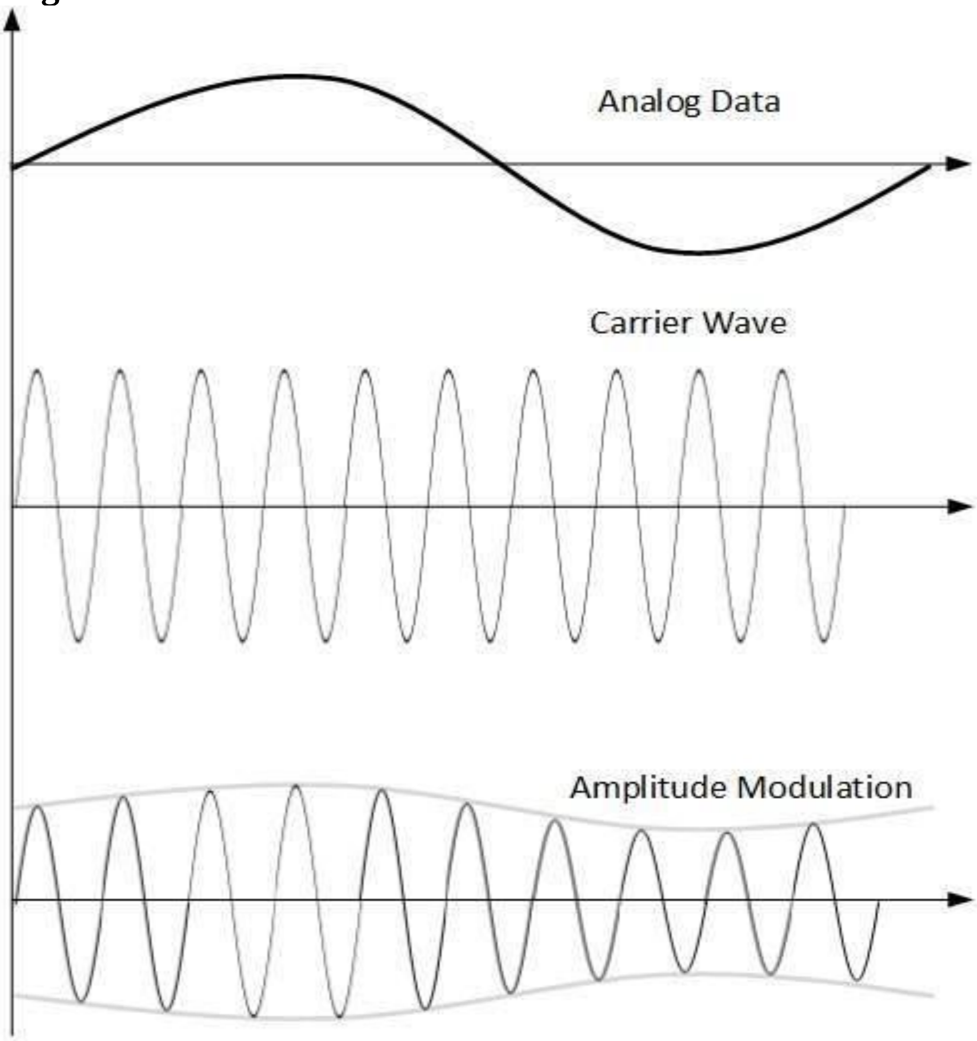
# Analog-to-Analog Conversion

**Analog signals are modified** to **represent analog data.** This conversion is also **known as Analog Modulation.** Analog modulation is **required when bandpass is used.** Analog to analog conversion can be **done in three ways.**




## Amplitude Modulation

In this modulation, the **amplitude of the carrier signal is modified to reflect the analog data.**



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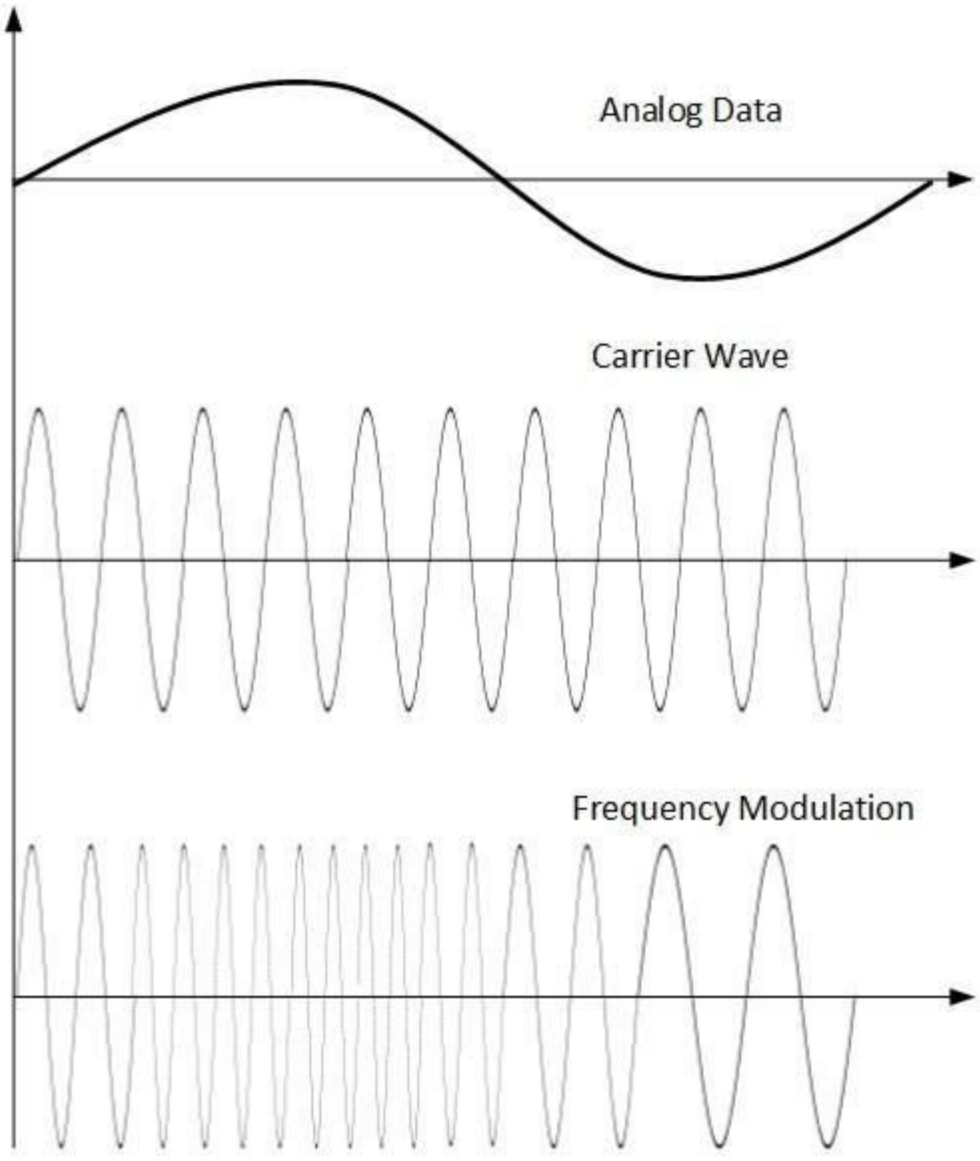
Amplitude modulation is **implemented by means of a multiplier.** The **amplitude of modulating signal (analog data) is multiplied by the amplitude of carrier frequency,** which then reflects analog data. The **frequency and phase** of carrier signal **remain unchanged.**

## Frequency Modulation

In this modulation technique, the **frequency of the carrier signal is modified to reflect the change in the voltage levels of the modulating signal (analog data).**




The **amplitude and phase of the carrier signal are not altered.**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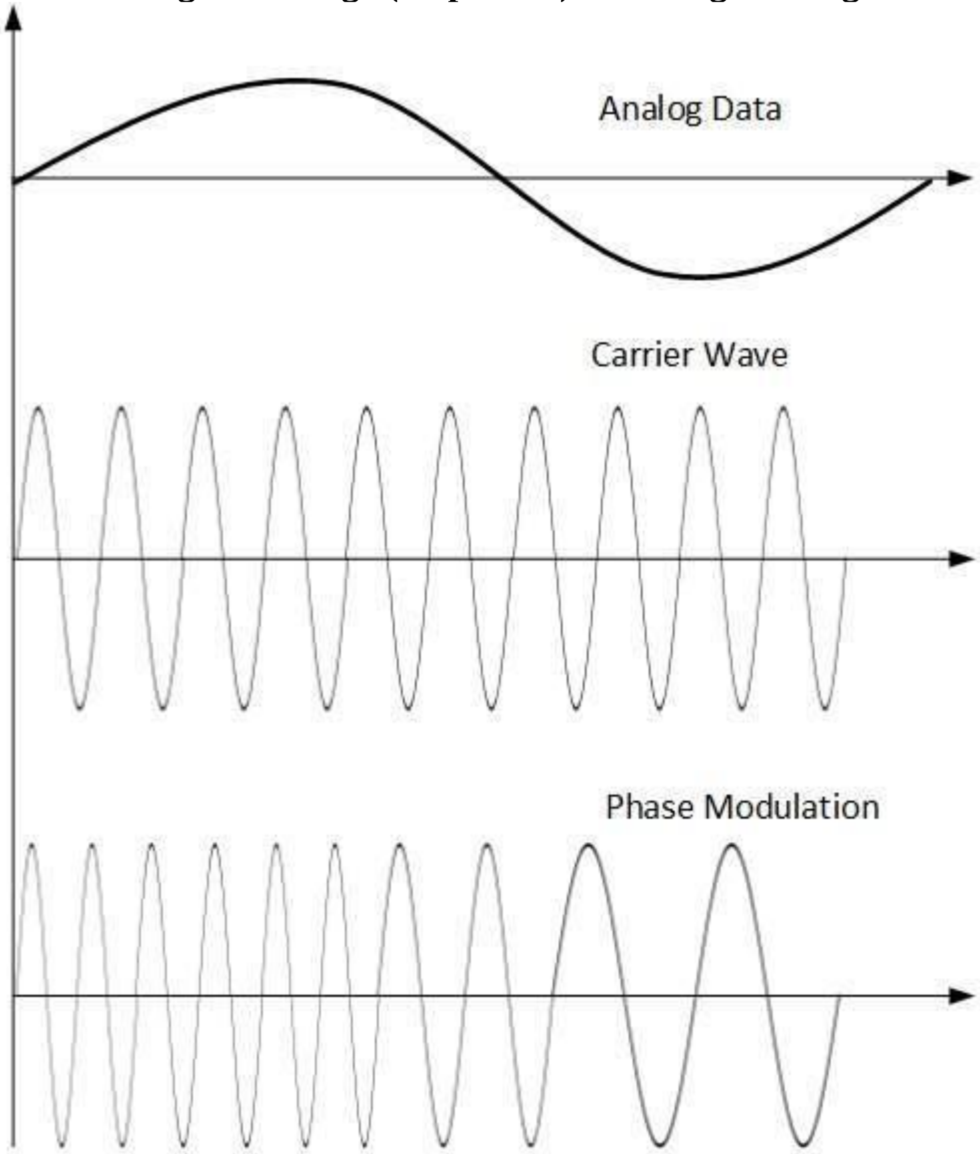
## Phase Modulation

In the modulation technique, the **phase of carrier signal is modulated in order to reflect the change in voltage (amplitude) of analog data signal.**




Phase modulation is practically **similar to Frequency Modulation,** but in Phase modulation **frequency of the carrier signal is not increased.** Frequency of carrier is signal is changed (made dense and sparse) to reflect voltage change in the amplitude of modulating signal.